Zoo **503 present paper**

Q1..island biogeography...2

Island biogeography is the study of the species composition and species richness on islands

Q2..nannoplankton..2

Defined as plankton less than 63 µm across, the smallest standard mesh size for sieves • Includes

organicwalled and siliceous forms

Q3..sexual dimorphism...3

dimorphism, in which males and females may show different sizes, and different specialized

features variations within species.

Q4 oldest fossil n their ages..5

The oldest eukaryote fossil may be Grypania, • A coiled, spaghetti-like organism that has been

reported from rocks as old as 1.85 Ga

The oldest locomotory trace fossils are from about 550 Ma northwest Russia • Fecal strings from

rocks some 600 Ma suggesting

Bacteria in rocks up to 3.2 Ga, indicated by stromatolites

Q5 .features of mollusca..5

The Phylum Mollusca can be traced back to at least the Late Precambrian, when Ki m b e r ell a

probably fed on algae in Ediacaran communities

Early Mollusks

Characterized by some short-lived, unusual forms • Molluskan features: mantle, mineralized

shell and radula • Members of the small shelly fauna.

6- Deuterostomes 5 marks

“Second mouth” however, have a mouth arising from a secondary opening • True blastopore

develops as anus • Echinoderm– Hemichordate Chordate (ppts)

Deuterostomia, (Greek: “second mouth”), group of animals—including those of thephyla

Echinodermata (e.g., starfish, sea urchins), Chordata (e.g., sea squirts,lancelets,

and vertebrates), Chaetognatha (e.g., arrowworms), and Brachiopoda(e.g., lamp

shells)—classified together on the basis of embryological development (google)

7 What is paleontology(2)

The branch of science concerned with fossil animals and plants. • the study of fossils to

determine organisms' evolution & ecology

8:. How anatomic filters helps to preserve complete fossil record(

Organisms are likely to be preserved only if they have hard parts, a skeleton of some kind. •

Entirely soft-bodied organisms, such as worms and jellyfish, are only preserved in rare cases.

9:What are the four characteristics of phylum Mollusca(5)

Unsegmented soft body with bilateral symmetry.

Presence of an internal or external shell.

A toothed tongue (made mostly of chitin) called the radula.

A mantle which is a fold in the body wall that lines the shell.

Muscular foot (and/or tentacles in some).

(copied from google)

10: What are the big five mass extinction events(5)

During mass extinctions, 20–90% of species were wiped out; these include a broad range of

organisms, and the events appear to have happened rapidly

It is difficult to study mass extinctions in the Precambrian • There have been a

Neoproterozoic event between the Ediacaran and Early Cambrian faunas.

The "big five" Phanerozoic mass extinctions occurred in the

• End-Ordovician

• Late Devonian

• End-Permian

• End-Triassic

• End-Cretaceous

11 :who introduce the concept of induction and explain it 2

Sir Francis Bacon • The enquirer might hope to see common patterns among the observations,

and these common patterns would point to an explanation, or law of nature. Induction

12): define speciation and write name of two models2

Speciation often occurs by the establishment of a barrier, and the isolation of part of a previously

interbreeding population.

“The process of splitting of a population to form two species is speciation ”.

• The allopatric • (“other homeland”) model

• The geographic model (based on the establishment of geographic barriers

13):write the name of animal groups who hits by end permian mass extinction event.3

The end-Permian mass extinction was the largest of all time • Probably caused massive volcanic

eruptions, leading to acid rain and global anoxia.(ppts)

During the period a number of animal groups became extinct, including the trilobites, tabulate

and rugose corals, and blastoids (see mass extinction). Amphibians and reptiles continued to be the dominant land animals and gymnosperms replaced ferns, clubmosses, and horsetails as

the dominant plants. "Permian. (google).

14 What is biostratigraphy 2m

Biostratigraphy is the establishment of fossil based successions and their use in stratigraphic

correlation • Biozones: The main operational units of a biostratigraphy.

15 : What is “Great Ordovician Biodiversification Event” (GOBE)? 5

Great Ordovician biodiversification event (GOBE), • Evolutionary radiation of animal life during

the Ordovician Period, 40 my after the CE.

Followed a series of Cambrian– Ordovician Extinction events • Fauna went on to dominate the

Palaeozoic • Marine diversity increased

Ordovician diversification generated witnessed a staggering increase in biodiversity at the

family, genus and species levels.

Some Mcqs are given below:

Solved MCCQs

Q1: Paleontology Originate from Greek word “Ontos” which means

(Creature)

Q2: Archaeopetyx is true missing link between

(Dinosaurs and Birds)

Q3: Trilobites dominated assemblages in England and Wales area were described by

(William Smith)

Q4: Fossils in Fold belts are, except

(In Excellent Shape)

Q5: Where an organism live matters in which of the filters is preservation of the complete fossil

recor

(Ecological Filters)

Q6: In Climate change in life history of earth has oscillated at least ---------- between icehouse

and greenhouse conditions

(5 times)

Q7: Scala Naturae is ----------

(The Ladder of life)

Q8: When local ecological conditions affect the form of an organism during its life time and

consequently fossil is called

(Ecophenotypic Effects)

Q9: Rohde and Muller (2005) argued that there is ---------- periodicity in mass extinctions

(62 myr)

Q10: Modification of the Oparin ---------- Haldane biochemical model is

(Hydrothermal Model)

Other MCQs..

Protozoan body plan...sponges..

Chromistan are except...protozoa..

Chelicerates are diverse and heterogeneous includes...a)mites .b)scorpian..c)spiderd)..all

Water vascular system in ...echinoderms

The group of gastropods...cenozoic

Fossil largest assemblagesof inadequate preserved fossils..

a) Fossil largersttaten are assemblage of?

b) vascular system is present in? echinoderms

c) chelicerate include? spiders,scorpions, all

d) acme group of gastropoda with neogastropods was in? paleozoic, mesozoic,

phanerozoic

e) belly-footed mollusks are?

f) parazoans body plan present in?

g) most advanced mollusks are?

b) Q1: Paleontology Originate from Greek word “Ontos” which means

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Other questions are following

Helicoidal

Some organisms are pentameral symmetry,

but the others were not

pentameral at all. One

such non-pentameral

group, the

helicoplacoids.

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